KAIZEN MATHS

# Lesson starter

**Cumulative Frequency Curves and Ogives**

Exam-Style Curves

Name _____

**SA1**

Complete the cumulative frequency column by adding the frequencies.

| Class interval | Frequency | Cumulative frequency |
|---|---|---|
| $20 \leq x < 25$ | 16 | ? |
| $25 \leq x < 30$ | 18 | ? |
| $30 \leq x < 35$ | 8 | ? |
| $35 \leq x < 50$ | 8 | ? |
| $50 \leq x < 65$ | 16 | ? |
| $65 \leq x < 75$ | 8 | ? |



Show your working

KAIZEN MATHS

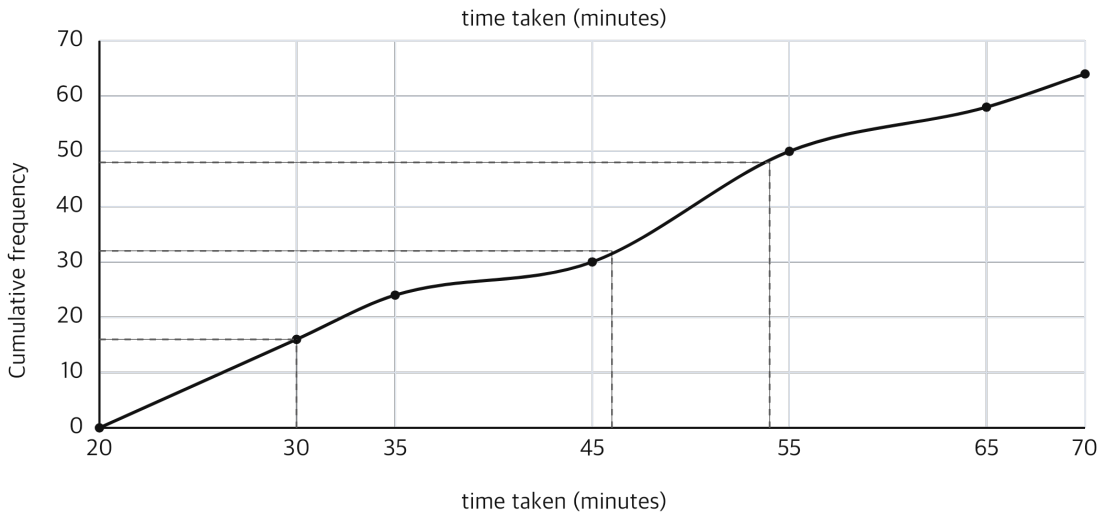
Lesson starter

**Cumulative Frequency Curves and Ogives**

Exam-Style Curves

Name _____

**SB1**




Estimate the five-number summary for a box plot.



Show your working

KAIZEN MATHS

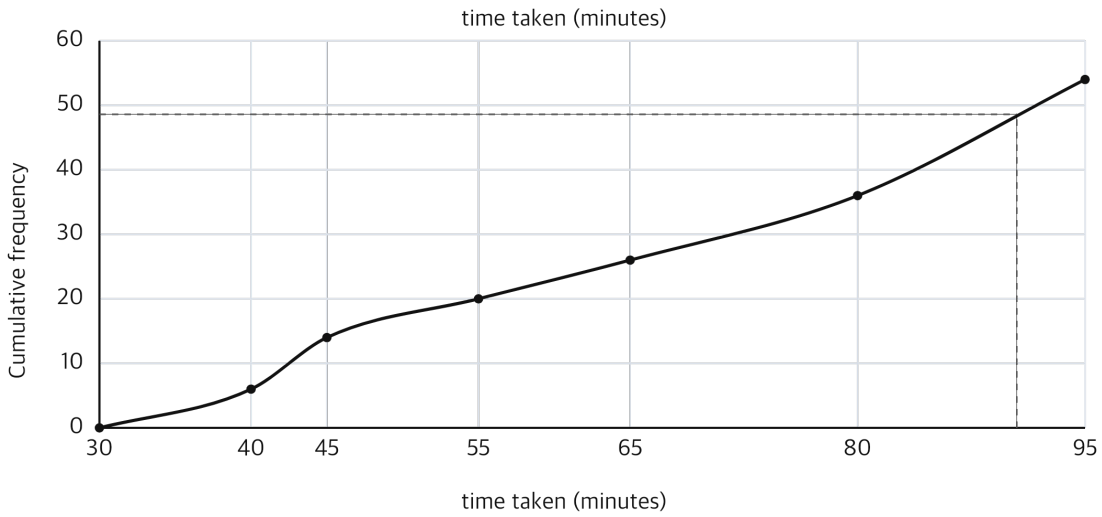
# Lesson starter

**Cumulative Frequency Curves and Ogives**

Exam-Style Curves

Name _____

**SC1**




Estimate the value that is exceeded by 10% of the students.

STA-5424-L3 · SC

Show your working

KAIZEN MATHS

# Lesson starter

**Cumulative Frequency Curves and Ogives**

Exam-Style Curves

Name _____

**SD1**

Complete the cumulative frequency column by adding the frequencies.

| Class interval | Frequency | Cumulative frequency |
|---|---|---|
| $10 \leq x < 20$ | 8 | ? |
| $20 \leq x < 35$ | 10 | ? |
| $35 \leq x < 50$ | 12 | ? |
| $50 \leq x < 65$ | 8 | ? |
| $65 \leq x < 75$ | 20 | ? |
| $75 \leq x < 90$ | 20 | ? |

STA-5424-L3 · SD

Show your working

KAIZEN MATHS

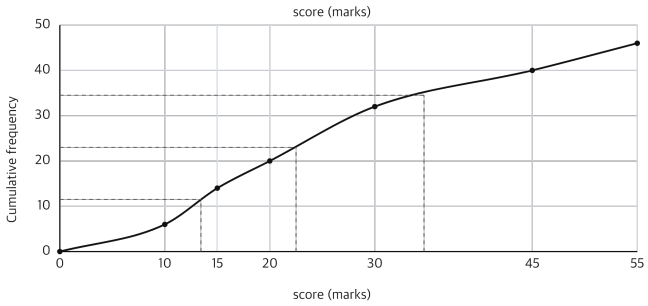
# Exit ticket

**Cumulative Frequency Curves and Ogives**

Exam-Style Curves

Name _____

**EA1**




Estimate the five-number summary for a box plot.



Show your working

KAIZEN MATHS

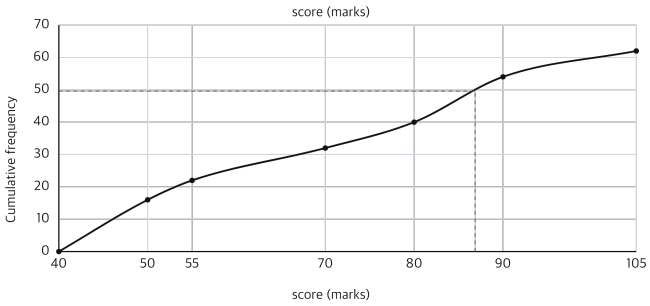
# Exit ticket

**Cumulative Frequency Curves and Ogives**

Exam-Style Curves

Name _____

**EB1**




Estimate the value that is exceeded by 20% of the candidates.



Show your working

KAIZEN MATHS

# Exit ticket

**Cumulative Frequency Curves and Ogives**

Exam-Style Curves

Name _____

**EC1**

Complete the cumulative frequency column by adding the frequencies.

| Class interval | Frequency | Cumulative frequency |
|---|---|---|
| $30 \leq x < 40$ | 6 | ? |
| $40 \leq x < 45$ | 12 | ? |
| $45 \leq x < 55$ | 12 | ? |
| $55 \leq x < 60$ | 20 | ? |
| $60 \leq x < 70$ | 18 | ? |
| $70 \leq x < 75$ | 20 | ? |



Show your working

KAIZEN MATHS

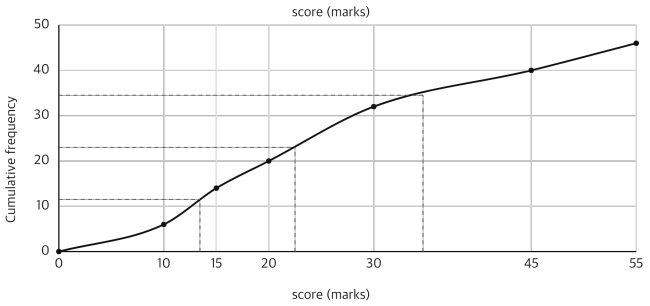
# Exit ticket

**Cumulative Frequency Curves and Ogives**

Exam-Style Curves

Name _____

**EA1**




Estimate the five-number summary for a box plot.



Show your working

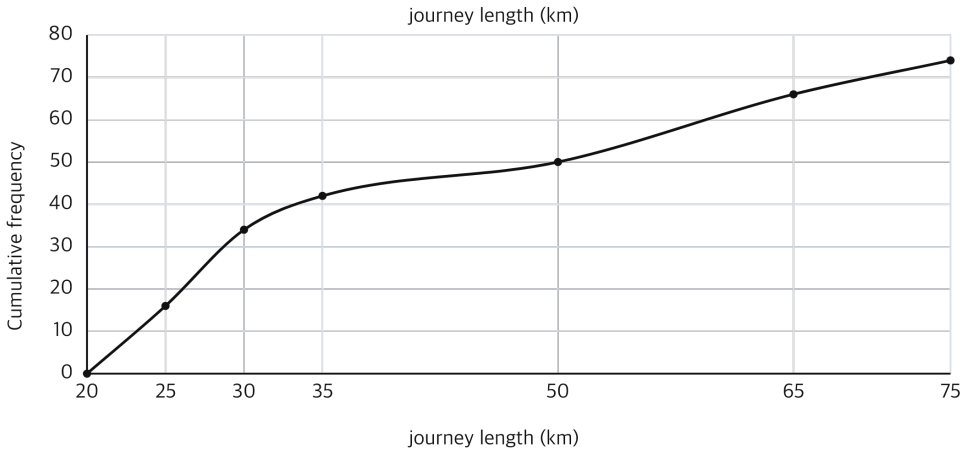
# Teacher answers & working

**Cumulative Frequency Curves and Ogives · Exam-Style Curves**

STA-5424-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

## SA1

Use the cumulative frequency curve to complete the cumulative frequency column.




| Class interval | Frequency | Cumulative frequency |
|---|---|---|
| $20 \leq x < 25$ | 16 | ? |
| $25 \leq x < 30$ | 18 | ? |
| $30 \leq x < 35$ | 8 | ? |
| $35 \leq x < 50$ | 8 | ? |
| $50 \leq x < 65$ | 16 | ? |
| $65 \leq x < 75$ | 8 | ? |

**Answer:**

20 ≤ x < 25 cumulative frequency 16; 25 ≤ x < 30 cumulative frequency 34; 30 ≤ x < 35 cumulative frequency 42; 35 ≤ x < 50 cumulative frequency 50; 50 ≤ x < 65 cumulative frequency 66; 65 ≤ x < 75 cumulative frequency 74

**Working:**

Read the cumulative frequency at each upper class boundary.

At x = 25, cumulative frequency ≈ 16

At x = 30, cumulative frequency ≈ 34

At x = 35, cumulative frequency ≈ 42

At x = 50, cumulative frequency ≈ 50

At x = 65, cumulative frequency ≈ 66

At x = 75, cumulative frequency ≈ 74

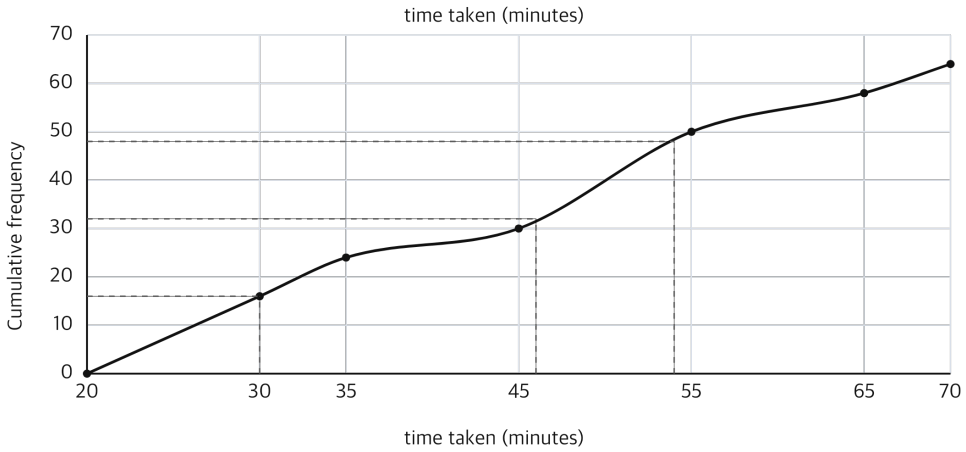
20 ≤ x < 25 cumulative frequency 16; 25 ≤ x < 30 cumulative frequency 34; 30 ≤ x < 35 cumulative frequency 42; 35 ≤ x < 50 cumulative frequency 50; 50 ≤ x < 65 cumulative frequency 66; 65 ≤ x < 75 cumulative frequency 74

# Teacher answers & working

**Cumulative Frequency Curves and Ogives · Exam-Style Curves**

STA-5424-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

## SB1




Estimate the five-number summary for a box plot.

**Answer:**

min 20, Q1 30, median 46, Q3 54, max 70

**Working:**

Use the minimum and maximum class boundaries for the endpoints.

minimum = 20, maximum = 70

Use the curve to read Q1, median, and Q3.

Q1 position = 16, so Q1 ≈ 30

median position = 32, so median ≈ 46

Q3 position = 48, so Q3 ≈ 54

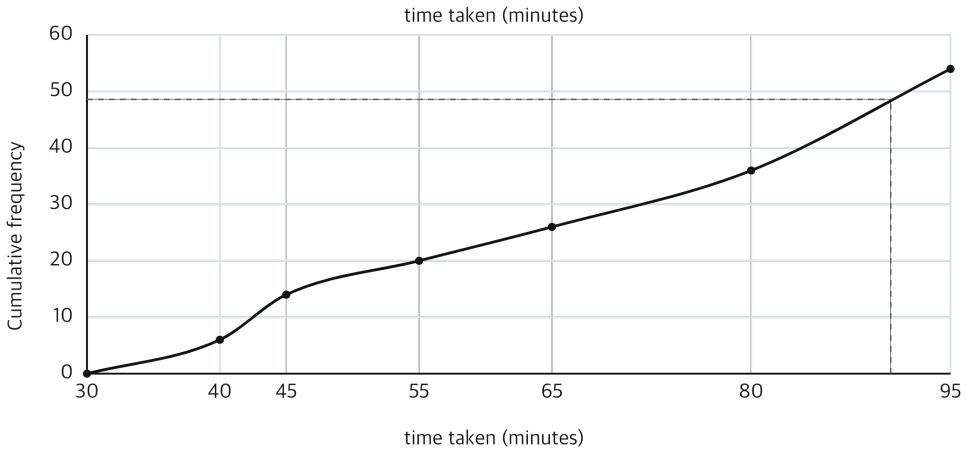
min 20, Q1 30, median 46, Q3 54, max 70

# Teacher answers & working

**Cumulative Frequency Curves and Ogives · Exam-Style Curves**

STA-5424-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

## SC1




Estimate the value that is exceeded by 10% of the students.

**Answer:**

90.5 minutes

**Working:**

If 10% exceed the value, then 90% are below it.

$90\%$ of $54 = 48.6$

Read the value from the curve at this cumulative frequency.

value ≈ 90.5 minutes

90.5 minutes

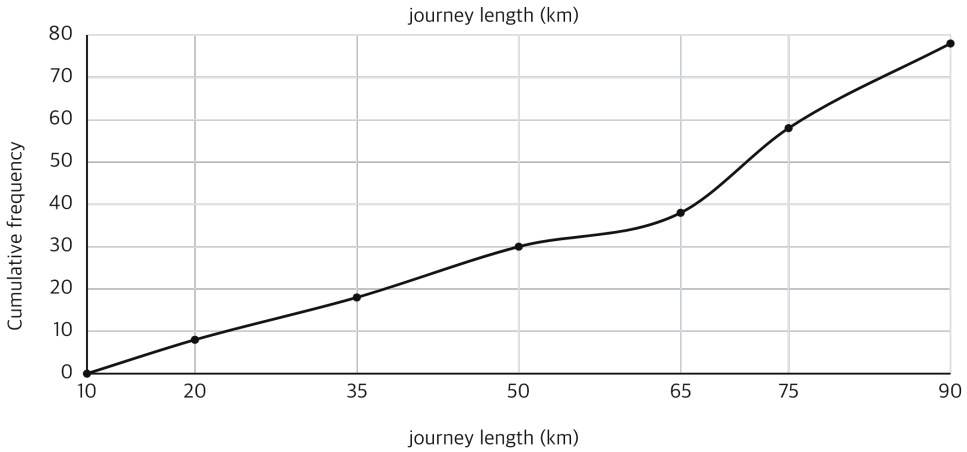
# Teacher answers & working

**Cumulative Frequency Curves and Ogives · Exam-Style Curves**

STA-5424-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

## SD1

Use the cumulative frequency curve to complete the cumulative frequency column.




| Class interval | Frequency | Cumulative frequency |
|---|---|---|
| $10 \leq x < 20$ | 8 | ? |
| $20 \leq x < 35$ | 10 | ? |
| $35 \leq x < 50$ | 12 | ? |
| $50 \leq x < 65$ | 8 | ? |
| $65 \leq x < 75$ | 20 | ? |
| $75 \leq x < 90$ | 20 | ? |

**Answer:**

10 ≤ x < 20 cumulative frequency 8; 20 ≤ x < 35 cumulative frequency 18; 35 ≤ x < 50 cumulative frequency 30; 50 ≤ x < 65 cumulative frequency 38; 65 ≤ x < 75 cumulative frequency 58; 75 ≤ x < 90 cumulative frequency 78

**Working:**

Read the cumulative frequency at each upper class boundary.

At x = 20, cumulative frequency ≈ 8

At x = 35, cumulative frequency ≈ 18

At x = 50, cumulative frequency ≈ 30

At x = 65, cumulative frequency ≈ 38

At x = 75, cumulative frequency ≈ 58

At x = 90, cumulative frequency ≈ 78

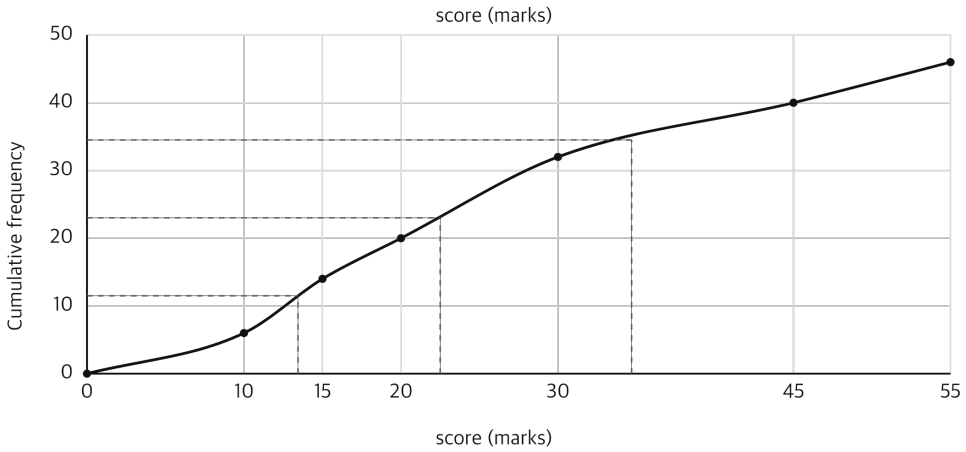
10 ≤ x < 20 cumulative frequency 8; 20 ≤ x < 35 cumulative frequency 18; 35 ≤ x < 50 cumulative frequency 30; 50 ≤ x < 65 cumulative frequency 38; 65 ≤ x < 75 cumulative frequency 58; 75 ≤ x < 90 cumulative frequency 78

# Teacher answers & working

**Cumulative Frequency Curves and Ogives · Exam-Style Curves**

STA-5424-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

## EA1




Estimate the five-number summary for a box plot.

**Answer:**

min 0, Q1 13.4, median 22.5, Q3 34.7, max 55

**Working:**

Use the minimum and maximum class boundaries for the endpoints.

minimum = 0, maximum = 55

Use the curve to read Q1, median, and Q3.

Q1 position = 11.5, so Q1 ≈ 13.4

median position = 23, so median ≈ 22.5

Q3 position = 34.5, so Q3 ≈ 34.7

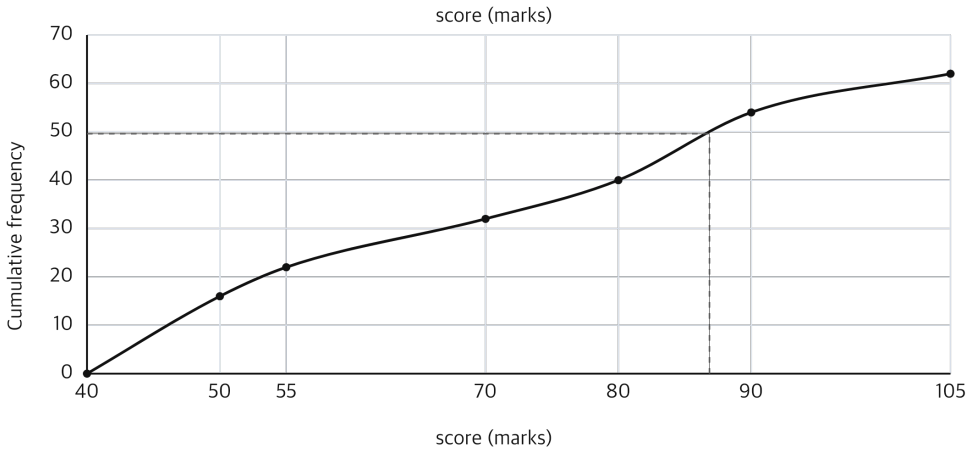
min 0, Q1 13.4, median 22.5, Q3 34.7, max 55

# Teacher answers & working

**Cumulative Frequency Curves and Ogives · Exam-Style Curves**

STA-5424-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

## EB1




Estimate the value that is exceeded by 20% of the candidates.

**Answer:**

86.9 marks

**Working:**

If 20% exceed the value, then 80% are below it.

$80\%$ of $62 = 49.6$

Read the value from the curve at this cumulative frequency.

value ≈ 86.9 marks

86.9 marks

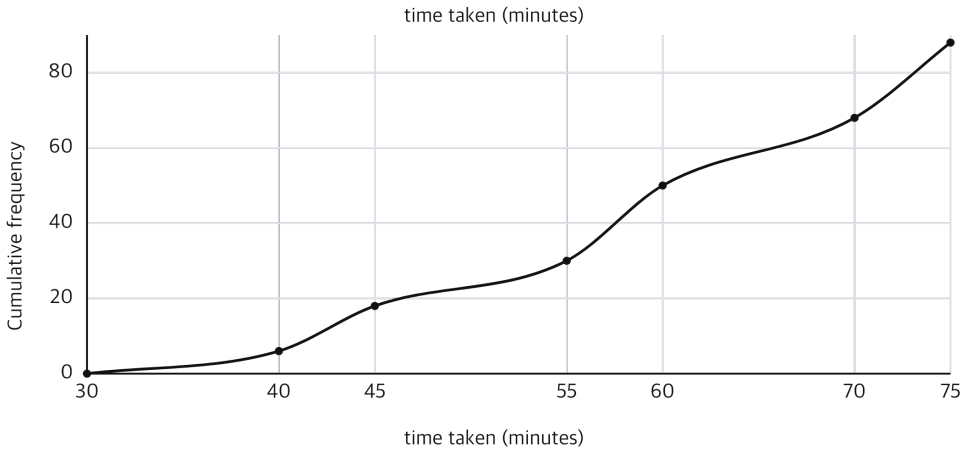
# Teacher answers & working

**Cumulative Frequency Curves and Ogives · Exam-Style Curves**

STA-5424-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

## EC1

Use the cumulative frequency curve to complete the cumulative frequency column.




| Class interval | Frequency | Cumulative frequency |
|---|---|---|
| $30 \leq x < 40$ | 6 | ? |
| $40 \leq x < 45$ | 12 | ? |
| $45 \leq x < 55$ | 12 | ? |
| $55 \leq x < 60$ | 20 | ? |
| $60 \leq x < 70$ | 18 | ? |
| $70 \leq x < 75$ | 20 | ? |

**Answer:**

30 ≤ x < 40 cumulative frequency 6; 40 ≤ x < 45 cumulative frequency 18; 45 ≤ x < 55 cumulative frequency 30; 55 ≤ x < 60 cumulative frequency 50; 60 ≤ x < 70 cumulative frequency 68; 70 ≤ x < 75 cumulative frequency 88

**Working:**

Read the cumulative frequency at each upper class boundary.

At x = 40, cumulative frequency ≈ 6

At x = 45, cumulative frequency ≈ 18

At x = 55, cumulative frequency ≈ 30

At x = 60, cumulative frequency ≈ 50

At x = 70, cumulative frequency ≈ 68

At x = 75, cumulative frequency ≈ 88

30 ≤ x < 40 cumulative frequency 6; 40 ≤ x < 45 cumulative frequency 18; 45 ≤ x < 55 cumulative frequency 30; 55 ≤ x < 60 cumulative frequency 50; 60 ≤ x < 70 cumulative frequency 68; 70 ≤ x < 75 cumulative frequency 88